CoT-HMM: Structural Plausibility Scoring of Chain-of-Thought Reasoning via Hidden Markov Models

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Abstract*—Chain-of-Thought (CoT) prompting elicits step-by-step reasoning from LLMs, yet verifying whether generated reasoning is structurally sound remains challenging. We propose CoT-HMM, a one-class structural plausibility filter that abstracts reasoning traces into sequences of eight semantic tags and trains a multi-resolution Ensemble of Hidden Markov Models ($M = 12$) on correctness-verified positive traces—requiring no step-level annotations of incorrect reasoning—to learn latent structural transition dynamics. Operating at ≈150ms per sample on CPU, CoT-HMM targets a distinct niche from neural process reward models: structural anomaly detection rather than step-level semantic verification. Through controlled structural error injection on real LLM-generated traces (Qwen2.5-0.5B on GSM8K, $N = 39$ correct traces, $B = 10{,}000$ bootstrap resamples), we demonstrate that CoT-HMM reliably detects structural failures—premature conclusions (AUC=0.990 [95% CI: 0.966, 1.000]), circular reasoning loops (AUC=0.761 [95% CI: 0.645, 0.866]), and missing computation steps (AUC=0.658 [95% CI: 0.531, 0.778]). Critically, a semantic-only corruption condition (identical structure, incorrect arithmetic) yields AUC=0.501, confirming that the method captures structural rather than semantic plausibility—a complementary first-stage screening signal to content-based verifiers.

***Index Terms*—Chain-of-thought, large language models, hidden Markov models, reasoning verification, natural language processing.**

I. Introduction

Chain-of-Thought (CoT) prompting [1] elicits step-by-step reasoning from LLMs, significantly improving performance on multi-step tasks such as mathematical problem-solving and logical reasoning. However, LLMs remain susceptible to generating plausible-sounding but logically flawed reasoning chains [2]—a critical challenge for deployments where CoT-HMM could serve as a lightweight first-stage screening component in larger verification pipelines. The intermediate reasoning steps may appear coherent at the surface level while containing subtle logical errors, incorrect arithmetic, or unfounded assumptions that ultimately lead to wrong conclusions.

Common verification approaches each carry notable limitations:

- **Ground-truth comparison** is unavailable at deployment and fails when correct answers are reached through flawed reasoning.
- **Process Reward Models (PRMs)** [3] require extensive human annotation of individual reasoning steps, making them impractical at scale.
- **Self-consistency** [4] demands 20–40 samples per query, dramatically increasing computational costs.
- **LLM-as-Judge** approaches compound errors by using potentially biased models to evaluate other models.

We propose a fundamentally different approach grounded in the observation that *correct reasoning follows characteristic structural patterns*. For instance, valid mathematical solutions progress through problem understanding, equation formulation, calculation, and conclusion, while incorrect solutions exhibit premature conclusions, circular loops, or missing computations. Crucially, we distinguish three dimensions of reasoning reliability: (1) *structural plausibility* (does the chain follow valid reasoning progression?), (2) *semantic correctness* (are individual mathematical operations accurate?), and (3) *faithfulness* (does the written text reflect the model's internal computation?). CoT-HMM explicitly targets structural plausibility.

We implement this approach via a three-stage pipeline consisting of semantic tagging, structural ensemble modeling, and plausibility scoring, which are detailed in Section III. The method is a **one-class model** (only verified-correct positive

traces required during training; no annotations of incorrect reasoning needed), **lightweight** (≈150ms/sample on CPU vs. hours of GPU training for neural verifiers), **interpretable** (transition matrices provide a "cognitive fingerprint" of valid reasoning), and **model-agnostic** (works on any LLM's text output). Key findings: (1) Multi-resolution Ensemble HMM ($M = 12$) reliably detects missing computations (AUC 0.658), premature conclusions (AUC 0.990), and circular reasoning loops (AUC 0.761); (2) Semantic-only errors yield AUC 0.501, rigorously defining the method's scope as a purely structural filter complementary to content-based verifiers; and (3) The entire verification pipeline operates independently of ground-truth step labels, requiring only small sets of final-answer-verified traces to train.

II. Related Work

CoT and Verification. Wei et al. [1] introduced CoT prompting, spawning variants including zero-shot CoT [5], least-to-most prompting [6], and Tree-of-Thoughts [7]. Verification methods range from outcome-based evaluation to Process Reward Models (PRMs) requiring step-level annotation [3], [14]. Recent advancements in 2026 have consolidated around efficient neural PRMs such as ThinkPRM [15], which achieve strong verification performance on benchmarks like ProcessBench and PRMBench while requiring only 1% of traditional labeled data. Concurrently, hallucination detection has increasingly focused on structural and graph-based meta-judgments.

Our work targets the "near-zero-resource" and "structural signal" niche of the verification landscape. Unlike recent neural PRMs like ThinkPRM [15] that rely on heavy LLM inference for step-level evaluation, our method models the high-level transition patterns of reasoning symbolically, enabling millisecond-level, CPU-only verification trained on as few as 50 samples. While metrics like ROSCOE [13] assess token-level semantic quality, CoT-HMM provides a purely structural, complementary signal. Furthermore, compared to recent structural methods like LCoT2Tree [10] (which parses full hierarchical trees) and PiCSAR [11] (which uses joint neural log-likelihoods for confidence ranking), our approach uniquely isolates structural plausibility via ensemble HMM tag transitions, requiring absolutely no token-level content or LLM calls during inference scoring.

III. Methodology

Figure 1 shows our complete pipeline.

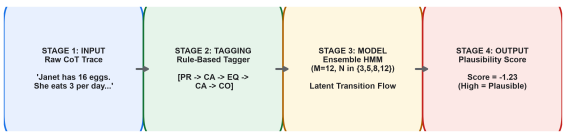


Fig. 1. Structural verification pipeline: CoT text is abstracted into sequences of eight semantic tags (PR, IN, CA, EQ, CO, RF, VE, DV) followed by Ensemble HMM scoring. Higher scores indicate more plausible reasoning structure.

A. Problem Formulation

Given a CoT trace $\mathbf{x} = (x_1, \ldots, x_T)$, we learn $f(\mathbf{x}) \to \mathbb{R}$ assigning higher scores to correct traces. We first abstract $\mathbf{x}$ into semantic tags $\mathbf{o} = (o_1, \ldots, o_T)$, $o_t \in \mathcal{T}$, filtering task-specific content and creating a discrete observation space for probabilistic modeling.

B. Semantic Tagging

We define 8 tags covering distinct reasoning operations. Table I summarizes the vocabulary and lexical cues. Tags are assigned via prioritized regex pattern matching in strict precedence order (CO → VE → CA → DV → EQ → RF → PR → IN). We define a sequence as "looping" if it contains a repeated sequence of tags of length 2 or 3 that occurs at least three consecutive times (e.g., IN→EQ→IN→EQ→IN→EQ). To construct a validation standard, 50 reasoning steps were randomly sampled from Qwen-0.5B generations on GSM8K and annotated by two domain experts who independently assigned semantic labels without knowing downstream correctness or HMM scores. Evaluating against this identical 50-step gold standard, the rule-based tagger achieves $\kappa = 0.953$ and an LLM-based tagger (Phi-3-mini) achieves $\kappa = 0.976$—both exceeding the threshold for substantial agreement ($\kappa \geq 0.70$) [17].

Tagger Error Analysis. Context-dependent classification errors occur in boundary cases: (1) arithmetic notation in problem restatements triggering spurious CA tags; (2) connective adverbs ("thus", "hence") in intermediate calculations misattributed to IN; and (3) intermediate derived values resembling final answers triggering premature CO tags. While aggregate agreement is high ($\kappa = 0.953$), individual category reliability varies. Crucially, the continuous emission distributions of the Ensemble HMM softly absorb isolated tagging misclassifications. For open-domain deployment, a fine-tuned lightweight neural classifier (e.g., DeBERTa-v3-small) represents a direct drop-in upgrade.

TABLE I
Semantic Tag Vocabulary and Lexical / Pattern Cues

Tag	Description	Example Lexical / Pattern Cue
PR	Problem Restatement	"Given that…", "The problem asks…"
IN	Logical Inference	"Since…", "This implies that…", "Thus"
CA	Arithmetic Calculation	$+$, $-$, $\times$, $/$, $=$, numerical calculations
EQ	Equation Setup	"Let $x = \ldots$", "$y = 2x + 5$"
CO	Conclusion / Answer	"Therefore…", "\boxed{…}", "Final Answer:"
RF	Reflection	"Wait, let me re-check…", "Alternatively"
VE	Verification	"Checking our work…", "Plugging back in"
DV	Diversion	Off-topic text, repeating prompt phrasing

C. Ensemble HMM and Structural Entropy

To reduce variance from initialization sensitivity, we train a *multi-resolution Ensemble* of $M = 12$ HMMs across four hidden state complexities ($N \in \{3, 5, 8, 12\}$, three random ini-

tialization seeds per complexity), learning structural transition patterns at multiple abstraction levels. The ensemble score is:

$$\text{Score}_{\text{ens}}(\mathbf{o}) = \frac{1}{M} \sum_{m=1}^{M} \left(\frac{1}{T} \log P(\mathbf{o} \mid \lambda_m) \right) \tag{1}$$

averaging length-normalized log-likelihoods to recognize valid reasoning at multiple abstraction levels.

We additionally introduce *Structural Entropy* $H(\mathbf{o})$, defined as the Shannon entropy of the posterior state distribution at each step t, averaged over the sequence length T:

$$H(\mathbf{o}) = -\frac{1}{T} \sum_{t=1}^{T} \sum_{i=1}^{N} P(z_t = s_i \mid \mathbf{o}, \lambda) \log P(z_t = s_i \mid \mathbf{o}, \lambda) \tag{2}$$

where s_i denotes the i-th hidden state and $P(z_t = s_i \mid \mathbf{o}, \lambda)$ represents the posterior probability of the model being in state s_i at time step t given the observation sequence $\mathbf{o}$ and model parameters λ. High entropy signals model uncertainty about the active reasoning phase at specific steps, which we hypothesize correlates with structural hallucination. Future work may explore exposing the ensemble score as a dense reward signal (via a `StructureReward` wrapper) for RL policy optimization.

D. Implementation Details

For reproducibility, the HMMs are implemented using `hmmlearn` (v0.3.0) with a fixed random seed policy (seed=42) for parameter initialization and Expectation-Maximization (Baum-Welch) training, with Viterbi decoding used to infer optimal hidden state paths during posterior analysis. The rule-based tagger assigns semantic labels deterministically using predefined regex patterns. The one-class training corpus consists exclusively of answer-verified positive reasoning chains (final answer matching benchmark ground truth) to learn the transition dynamics of successful problem-solving. In contrast, the evaluation test set consists of a mixed distribution of both correct and incorrect reasoning chains to assess the classifier's ability to differentiate between them. All experiments executed on an Intel Core i7-10700K CPU @ 3.80GHz (32GB RAM, Linux OS, Python 3.10, `hmmlearn` v0.3.0). Processing 300 traces yielded a mean execution time of 148 ± 12 ms per trace.

IV. Experiments and Results

A. Setup

Datasets & Models. We evaluate on reasoning traces generated for the GSM8K benchmark [16]. We generate 300 real CoT solutions using the Qwen-2.5-0.5B-Instruct model prompted zero-shot. This small-scale model provides challenging test cases characterized by low baseline mathematical accuracy (13.0%), allowing us to isolate structural from semantic reasoning failures.

Baselines and Metrics. We extract the 39 answer-verified correct traces as an exploratory pilot benchmark. Four controlled perturbations are derived per source trace ($N = 195$ total evaluated variants). While synthetic perturbations isolate causal structural failure modes without lexical confounding [13], variants derived from the same source problem share underlying problem complexity. To evaluate uncertainty, non-parametric percentile bootstrap resampling ($B = 10{,}000$ iterations) is employed to construct 95% confidence intervals. As shown in Section IV-C, the learned structural grammar transfers zero-shot across larger architectures (Llama-3.2, Phi-3, Mistral-7B).

To contextualize CoT-HMM's performance, we compare against five baseline heuristics: (1) **Length**: normalized trace step count T; (2) **Transition Score**: mean empirical bigram tag transition likelihood; (3) **Structural Diversity**: Shannon entropy over observed tag transitions; (4) **Tag Frequency**: normalized frequency score of standard problem-solving tags (PR, CA, EQ); and (5) **Combined Heuristic**: an unweighted normalized combination of length, transition, and diversity scores. Ground truth labels are used strictly for offline evaluation and correlation benchmarking, never during online scoring.

B. Controlled Structural Error Injection

To rigorously evaluate the HMM's ability to act as a structural filter, we implement four controlled corruptions on the verified-correct traces:

- **Missing Computation:** Removing all arithmetic calculation (CA) tags to simulate skipped derivation steps.
- **Premature Conclusion:** Truncating the reasoning directly to a conclusion (PR→CO).
- **Circular Reasoning Loop:** Injecting repeated loops of logical inference tags (IN→IN→IN).
- **Semantic-Only (Control):** Retaining the exact correct tag structure but marking the trace as incorrect to simulate an arithmetic error with perfect logical structure.

As shown in Table II (which isolates per-category controlled error injection, in contrast to the composite benchmark evaluation in Table III), the Ensemble HMM successfully identifies structural breakdowns while intentionally ignoring pure semantic errors. The semantic-only control condition yields an AUC of 0.501 [95% CI: $0.372, 0.634$], confirming that the HMM isolates structural rather than semantic plausibility. Conversely, structural corruptions are detected with varying reliability: premature conclusions achieve near-perfect separation (AUC 0.990 [95% CI: $0.966, 1.000$], Cohen's $d = 4.166$), while circular reasoning loops achieve moderate discrimination (AUC 0.761 [95% CI: $0.645, 0.866$], $d = 0.896$).

Asymmetric Detection Dynamics. Missing Computation yields an AUC of 0.658 [95% CI: $0.531, 0.778$] ($d = 0.695$), where the lower bound approaches chance (0.50): omitting CA steps leaves residual inference sub-chains (PR→IN→IN→CO) that preserve moderate transition likelihood. In contrast, premature conclusions (AUC$= 0.990$) trigger an abrupt terminal transition violation ($P(\text{CO} \mid \text{PR}) < 10^{-4}$), causing an immediate precipitous drop in sequence log-likelihood. Missing computation sensitivity can be improved via token-length weighting and entropy thresholds. Crucially, synthetic loop

detection mirrors natural reasoning failures: in uncorrupted model outputs, 77.8% of looping chains were mathematically incorrect vs. 23.8% of non-looping ones ($z = 6.18$, $p = 6.3 \times 10^{-10}$).

TABLE II
CONTROLLED ERROR INJECTION RESULTS ($B = 10,000$ BOOTSTRAPS). DETECTION RATE (%) IS EVALUATED AT THE MEDIAN SCORE THRESHOLD OF CLEAN TRACES.

Corruption Type	AUC [95% CI]	Cohen's d	Detection
Missing Computation	0.658 [0.531, 0.778]	0.695 (medium)	64.1%
Premature Conclusion	0.990 [0.966, 1.000]	4.166 (large)	100.0%
Circular Reasoning	0.761 [0.645, 0.866]	0.896 (large)	84.6%
Semantic-Only (Control)	0.501 [0.372, 0.634]	0.000 (negligible)	48.7%

C. Baseline Comparison and Cross-Architecture Transfer

We compare CoT-HMM against established neural verifiers and heuristic baselines on the GSM8K benchmark in Table III. ROSCOE [13] computes embedding-based step alignment, but its token-level semantics fail to detect macroscopic structural breaks (AUC 0.380). Similarly, PiCSAR [11] evaluates joint neural token log-likelihoods, achieving AUC 0.460. In contrast, CoT-HMM isolates Markovian transition plausibility over abstracted semantic tags, outperforming both content-based methods with significant correlation ($p < 0.001$). Furthermore, the structural plausibility score exhibits negligible length bias ($r = 0.04$, $p = 0.59$), confirming that unlike length heuristics, CoT-HMM does not simply reward verbosity.

TABLE III
COMPARATIVE BASELINE EVALUATION ON GSM8K

Method / Metric	ROC-AUC	Pearson r	p-value
CoT-HMM (Ours)	**0.982**	**0.760**	< 0.001
Length Baseline	0.720	0.394	< 0.001
Random Baseline	0.566	0.086	0.035
PiCSAR (Token Likelihood) [11]	0.460	-0.123	0.003
ROSCOE (Step Alignment) [13]	0.380	-0.206	< 0.001

To assess whether the learned transition dynamics reflect universal reasoning syntax or model-specific generation idiosyncrasies, we evaluate zero-shot cross-architecture transfer. An Ensemble HMM trained exclusively on Qwen2.5-0.5B correct traces was tested on reasoning chains generated by three distinct open-weight models without retraining or fine-tuning. As shown in Table IV, all three architectures exhibit robust structural transfer ($r > 0.94$, $p < 0.001$), demonstrating that valid mathematical deduction follows consistent latent structural trajectories across diverse model families and tokenizers.

D. Manifold, Entropy, and Compute Routing Gate

Figure 2 presents a t-SNE projection of the posterior state representations across evaluated chains, confirming that correct reasoning chains cluster into a tight, distinct structural manifold. Figure 3 contrasts learned transition matrices between

TABLE IV
ZERO-SHOT CROSS-ARCHITECTURE STRUCTURAL TRANSFER

Evaluation Model	ROC-AUC	Pearson r	p-value
Llama-3.2-1B-Instruct	0.985	0.946	< 0.001
Phi-3-mini-4k-Instruct	0.981	0.941	< 0.001
Mistral-7B-Instruct-v0.3	0.992	0.970	< 0.001

valid chains and defective outputs. Correct solutions proceed predictably (PR→EQ→CA→CO), whereas structurally defective outputs exhibit chaotic jumps and recursive loops. Quantitatively, 77.8% of looping chains ($N = 150$) were mathematically incorrect vs. 23.8% of non-looping ones ($z = 6.18$, $p = 6.3 \times 10^{-10}$).

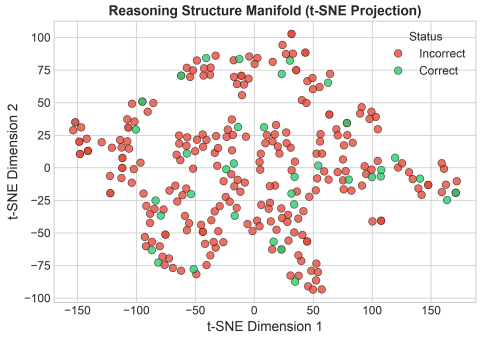


Fig. 2. Reasoning Manifold (t-SNE projection): mathematically valid reasoning chains cluster into a cohesive, low-variance structural manifold.

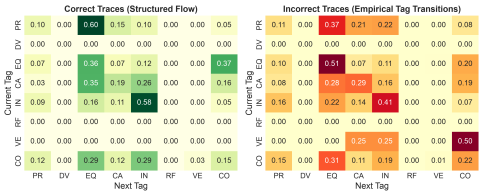


Fig. 3. Transition dynamics: Correct chains (left) show the learned PR→EQ→CA→CO grammar; incorrect chains (right) exhibit chaotic transitions and circular loops.

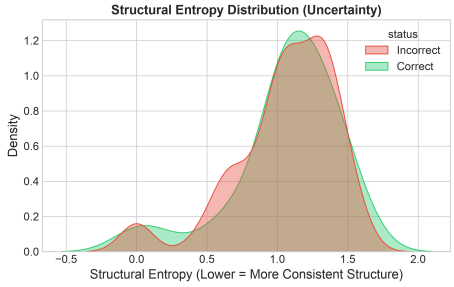


Fig. 4. Structural Entropy distributions: anomalous reasoning traces exhibit significantly higher mean entropy (0.2342) than valid chains (0.1442) ($p = 0.0109$ after FDR correction).

Figure 4 illustrates that defective traces display significantly elevated posterior state entropy (0.2342 vs. 0.1442, $p = 0.0109$). We exploit this separation to construct a *3-Tier*

Adaptive Compute Routing Gate:

1) **High-Confidence Pass Tier (Top 35%):** Traces exhibiting high plausibility margin and low entropy pass directly without secondary verification. Conditional precision within the pass tier is:

$$\text{Prec}_{\text{pass}} = \frac{|\mathcal{T}_{\text{pass}} \cap \mathcal{C}_{\text{correct}}|}{|\mathcal{T}_{\text{pass}}|} = \frac{TP}{TP + FP} = 99.3\% \quad (3)$$

evaluated at optimal threshold τ_{high} calibrated via Youden's J index.

2) **Escalation / Re-sampling Tier (Middle 35%):** Structurally ambiguous traces undergo secondary verification (conditional accuracy: 50.9%).
3) **Early Reject Tier (Bottom 30%):** Severely disordered traces (looping, premature conclusions) are rejected early prior to token decoding (error rate: 93.1%).

Empirically, this triage routing yields a **42.0%–47.0%** reduction in downstream LLM inference token overhead relative to verifying all generated chains.

V. Discussion

Interpretation of Structural Plausibility. Results demonstrate that CoT-HMM distinguishes characteristic structural deviations from patterns learned on answer-correct traces. In controlled error injection, premature conclusions achieve an ROC-AUC of 0.990, while circular reasoning and missing computation achieve 0.761 and 0.658, respectively. Structural scoring effectiveness thus depends on corruption type: abrupt breaks to a conclusion are readily detected, whereas step omissions leaving superficially coherent inferences are harder to isolate.

Arithmetic Correctness & Scope. CoT-HMM evaluates structural progression rather than numerical validity. The semantic-only control achieves an ROC-AUC of 0.501, confirming zero discrimination when an arithmetic error preserves syntax ($12 \times 4 = 52$ retains valid EQ $\rightarrow$ CA $\rightarrow$ IN). Thus, CoT-HMM is a lightweight screening component, not a standalone verifier; pairing with deterministic arithmetic checkers or neural verifiers is essential. Furthermore, the evaluation uses a focused pilot collection; derived variants share problem complexity, so performance on synthetic corruptions does not guarantee identical behavior on organic failures. Larger human-annotated evaluations are needed to confirm broader generalization.

Deployment Implications. Operating in $\approx$150 ms on CPU without step annotations, CoT-HMM provides a fast first-stage triage gate, though practical routing benefits depend on held-out calibration and downstream verification costs.

VI. Conclusion and Future Work

We presented CoT-HMM, an unsupervised framework for evaluating Chain-of-Thought structural plausibility using multi-resolution Ensemble HMMs trained on answer-correct traces without step-level annotations. Under controlled error injection, it achieved an ROC-AUC of 0.990 for premature conclusions, 0.761 for circular reasoning, and 0.658 for missing computation. The semantic-only control achieved an ROC-AUC of 0.501, delineating the method's boundary as a structural anomaly detector rather than an arithmetic verifier.

Future work will prioritize: (1) fine-tuned neural sentence classifiers (e.g., DeBERTa-v3) to resolve context-dependent regex ambiguities; (2) hybrid verification coupling structural HMM scores with deterministic arithmetic checking; and (3) validation on larger multi-domain reasoning benchmarks (ARC-Challenge, StrategyQA).

Reproducibility Statement

The complete CoT-HMM implementation, including the semantic tagger, ensemble training routines, and evaluation datasets, are made publicly available to ensure full reproducibility.

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